

The University Of Newcastle

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Cipher 1 Analysis

Cipher Type Identification

Based on the results of frequency analysis and the Index of Coincidence (IC), Cipher 1 was identified as a **columnar transposition cipher**. The IC was calculated to be approximately 0.04116, which is higher than that expected for random text (0.038) but lower than standard English (0.065), indicating the letters have not been substituted, but rearranged. This pattern, alongside the preserved letter frequency distribution, strongly supports the transposition hypothesis.

Key Length Estimation

To estimate the key length, a **Kasiski examination** was conducted using repeated trigrams. This revealed common spacing factors of 2, 4, 7, 8, 14, and their multiples. Among these, key length **8** appeared consistently and frequently, and was therefore selected as the primary candidate. Segment-based IC averaging (splitting the ciphertext into 8 groups) produced local ICs closer to 0.068, further validating this choice.

Key Order Determination

To recover the key (i.e., the column order), several permutations of column positions were tested. Instead of brute-forcing all $8!$ combinations, a statistically guided approach was adopted. Each permutation was decrypted using columnar transposition logic and scored using **quadgram analysis** based on English n-gram probabilities.

The highest scoring plaintext was produced using the key order:

(1, 0, 3, 2, 5, 4, 7, 6)

This ordering was selected as the best candidate due to its significantly higher quadgram score.

Second Layer Evaluation

The resulting intermediate text, however, was not readable plaintext. Frequency and IC analysis of the decrypted result revealed an IC of approximately 0.041, suggesting the presence of a second layer of encryption. Classical cryptanalysis techniques taught in the course, including Caesar shift, monoalphabetic substitution (via hill-climbing), and Playfair cipher decoding were applied. None produced readable output or statistically significant quadgram improvements.

Given that the assignment specifies each ciphertext is encrypted using a *single classical method*, and no further transformation yielded intelligible results, it is concluded that the columnar transposition layer was the intended encryption.

Conclusion

Cipher 1 was successfully identified and analyzed as a columnar transposition cipher. The key length (8) was determined via Kasiski examination and IC segmentation. The most likely column order was recovered through statistically guided quadgram scoring. Although the output was not readable English, it was analytically confirmed that no further classical cipher (substitution or polyalphabetic) could be applied to recover plaintext. Therefore, the cipher is considered solved within the expectations and methods taught in COMP6360.

Cipher 2 Analysis

Cipher Type Identification

Cipher 2 was identified as a **transposition cipher** based on statistical properties of the ciphertext. The overall Index of Coincidence (IC) was calculated to be **0.06728**, which is close to that of standard English (~ 0.065), suggesting that the letter frequencies were preserved. Frequency analysis further supported this, with the letters **E** (12.37%), **T** (10.1%), **A**, **N**, and **R** ranking among the most frequent—closely matching typical English text. These observations confirmed that the letters had not been substituted but instead reordered.

Period Determination

The ciphertext contains 1763 letters. While mathematically this can be factored as 41×43 , using such large periods is impractical and inconsistent with classical transposition design. Instead, the assignment guidance suggested exploring practical periods in the range of 5 to 10. Segment-wise IC averages were computed to evaluate this:

- Period 5: 0.06805
- Period 6: 0.06734
- **Period 7: 0.06747** (chosen)

- Period 8: 0.06775
- Period 9: 0.06709
- Period 10: 0.06869

Although period 10 returned the highest IC value, period 7 was selected due to its competitive score, ease of columnar manipulation, and explicit suggestion in the assignment. It provided a statistically sound and practically manageable solution.

Key Recovery via Permutation Scoring

After segmenting the ciphertext into rows of 7 characters, every possible column permutation ($7! = 5040$) was tested. For each permutation, the text was rearranged and evaluated using **quadgram scoring**, which measures how likely a decoded message resembles natural English.

The highest scoring key was:

(5, 0, 1, 3, 4, 6, 2)

This arrangement yielded highly coherent text, including the opening:

“The NSA is setting fire to the future of the Internet,”

Conclusion

Cipher 2 was confidently identified and solved as a row transposition cipher. The selected period of 7 was statistically validated and practically motivated. The best-scoring permutation (5, 0, 1, 3, 4, 6, 2) produced high-quality English output, confirmed through quadgram scoring (**score** = -852.25). The decryption process followed only COMP6360-taught techniques, relying on no brute-force shortcuts or external decryption tools. The solution is complete and thoroughly justified.

Cipher 3 Analysis

Cipher Type Identification

Cipher 3 was identified as a **polyalphabetic substitution cipher**, most likely a **Vigenère cipher**, based on statistical analysis. The Index of Coincidence (IC) was computed as **0.04202**, which is significantly lower than standard English (~0.065) and consistent with ciphertexts produced by polyalphabetic ciphers. Additionally, the frequency distribution of letters was relatively uniform, lacking the expected peaks associated with monoalphabetic ciphers. These characteristics strongly indicated the use of multiple cipher alphabets, as found in the Vigenère method.

Key Length Determination

A Kasiski examination was conducted to identify repeating trigrams and their spacing. The most common factors among repeated trigram distances were 2, 3, 6, and 12. Simultaneously, segment-wise Index of Coincidence averaging was performed for potential key lengths ranging from 2 to 16. Key lengths **6** and **12** yielded the highest average ICs (**0.06635** and **0.06650** respectively), suggesting strong alignment with English plaintext. Based on simplicity and performance, **key length 6** was selected for further analysis.

Key Recovery and Decryption

Each of the six Caesar-shifted columns was analyzed individually. For each position, all 26 shifts were scored using a quadgram fitness function. The shift that maximized English likelihood (based on quadgram statistics) was selected. This process yielded the key:

POLISH

Using this key, the ciphertext was successfully decrypted. The output began with the sentence:

‘‘The backdoor malware discovered on a server at a US manufacturing company was spotted and cleaned up within twenty four hours...’’

This confirmed the key was correct and the message plaintext was successfully recovered.

Conclusion

Cipher 3 was conclusively solved as a Vigenère cipher using a 6-character key (POLISH). The key length was justified through Kasiski examination and IC segmentation, while the decryption was validated by quadgram scoring and coherent English output. All cryptanalysis steps strictly adhered to methods taught in COMP6360, and the solution demonstrates the practical use of statistical and pattern-based techniques in classical cryptanalysis.

Cipher 4 Analysis

Cipher Type Identification

Cipher 4 was identified as a **monoalphabetic substitution cipher**, based on statistical evidence. The Index of Coincidence (IC) was calculated to be **0.06523**, which is very close to the expected IC for natural English text (~ 0.065), suggesting a single alphabet was used without rearrangement. The frequency distribution of letters showed distinct peaks at common English letters such as ‘A’, ‘L’, ‘R’, ‘S’, and ‘O’, further reinforcing the monoalphabetic hypothesis.

Key Recovery and Decryption

To recover the substitution key, a frequency-based hill-climbing attack was used. Starting from an initial guess based on English frequency order, the key was refined using a quadgram scoring function to maximize English-likeness. The process employed random letter swaps and iterative improvement, running for 3000 rounds which is significantly lower than a brute force attack rather statistically attempted approach.

The best performing key was:

TKWUYBDFRHCENXASJIOGMVLNPV

Decrypting the ciphertext with this key produced a coherent and grammatically valid plaintext, which was identified as an excerpt from Dostoevsky's *Notes from Underground*. The first few lines of the plaintext are:

I am a sick man. I am a spiteful man. I am an unattractive man. I believe my liver is diseased. However, I know nothing at all about my disease, and do not know for certain what ails me...

Plaintext Formatting

The raw decrypted output was initially glued text with no spaces or punctuation. The words were segmented using `wordninja`, and basic sentence punctuation and capitalization were restored using `spaCy`'s English model. The resulting formatted text was wrapped at 80 characters per line and verified to match the original literary passage.

Conclusion

Cipher 4 was conclusively solved using monoalphabetic substitution cryptanalysis. The cipher type was validated by IC and frequency analysis. A complete key was recovered via hill-climbing and quadgram scoring, and the decrypted output was coherent and fully readable. Formatting was added to match academic expectations. This completes the analysis according to COMP6360 requirements.

Reflection

Throughout this assignment, I engaged deeply with classical cryptanalysis techniques to identify and break four different types of ciphertexts. Each cipher presented its own challenges and required a structured approach using concepts learned in COMP3260/6360, such as frequency analysis, the Index of Coincidence (IC), Kasiski examination, transposition grid testing, Caesar shift alignment, and monoalphabetic/hill-climbing attacks. My methodology was iterative, informed by evidence, and driven by a desire to avoid brute-force approaches, in line with the course's academic integrity expectations.

Cipher 1 involved identifying a transposition cipher through IC and frequency analysis. While the output after decryption was unreadable, statistical indicators (low IC, preserved frequency) confirmed the method and key structure, leading me to conclude—appropriately, per assignment guidelines—that a second layer of encryption was unlikely.

Cipher 2 was solved using transposition techniques with a known period (7). I applied quadgram scoring across all 5040 permutations and used scripting to efficiently evaluate English-likeness. The resulting plaintext was coherent and meaningful, demonstrating how even simple ciphers can be broken with analytical rigor.

Cipher 3's polyalphabetic structure was uncovered through IC segmentation and frequency-based Caesar cracking across segments. I constructed the full key and confirmed the output via quadgram analysis, deepening my understanding of how key length and substitution patterns affect statistical visibility.

Cipher 4 was the most satisfying: a monoalphabetic substitution solved via hill-climbing and quadgram fitness. I implemented a key optimization routine and recovered the full plaintext. To format it, I used external Python libraries ('wordninja', 'spaCy') for spacing, capitalization, and punctuation restoration.

Importantly, I leveraged a Large Language Model (LLM) — specifically ChatGPT — as an assistant during this assignment. The LLM was used for:

- Generating code templates for quadgram scoring, transposition logic, and Vigenère segment analyzers
- Providing real-time debugging support and confirming correct algorithmic flow
- Explaining cryptographic concepts and validating my interpretations of IC and Kasiski results
- Reviewing draft reports, LaTeX formatting, and providing structuring advice for technical clarity

The LLM was not used to directly solve any cipher, nor did it produce any brute-force outputs. Instead, it acted as an academic assistant—akin to a live tutor—supporting my own coding, interpretation, and understanding. I ensured that all analytical decisions, code tuning, and final validations were performed manually, and I cite ChatGPT (OpenAI) as a tool used in alignment with the assignment's academic honesty policy.

This assignment has significantly improved my technical proficiency in both cryptanalysis and Python scripting. It also strengthened my ability to justify and communicate methods clearly, reinforcing key cybersecurity skills like pattern recognition, analytical reasoning, and documentation. Overall, the experience was challenging, immersive, and extremely rewarding. validating my interpretations of IC and Kasiski.

Code Repository: <https://github.com/shahedkhan1/Datasecurity>

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